Name: _____ Date: _____ Class: _____

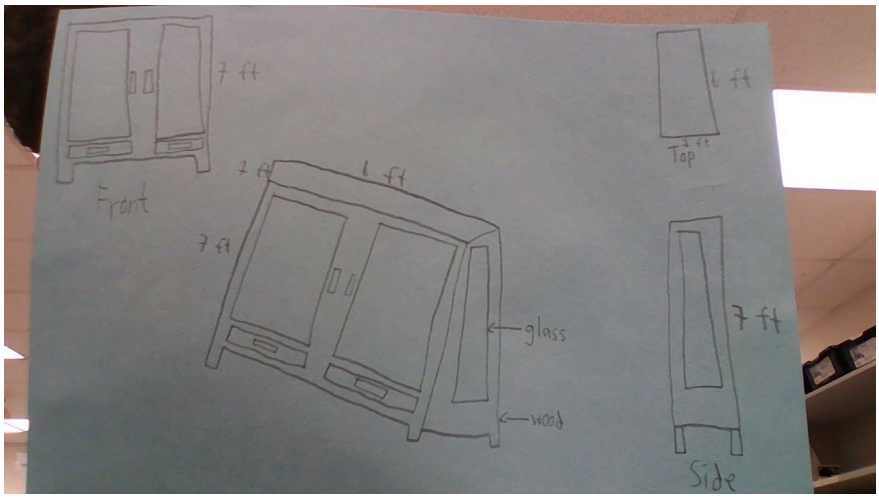
MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea #1

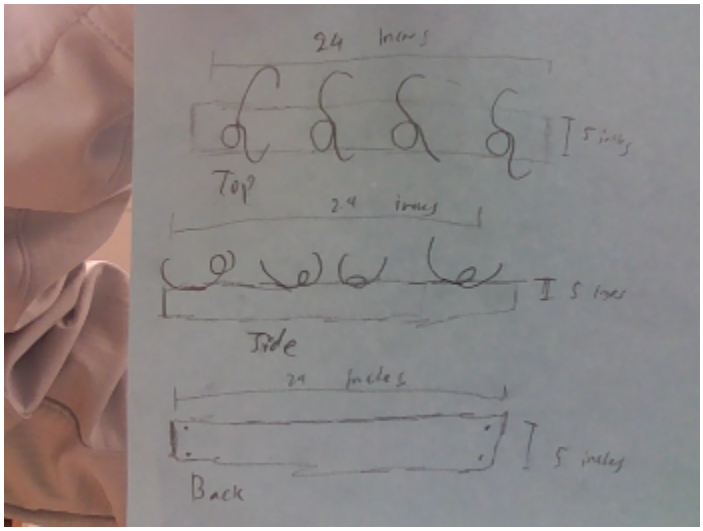
Date: 03/05/2025
Description: Create a closet with cabinets for each individual guitar.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #2

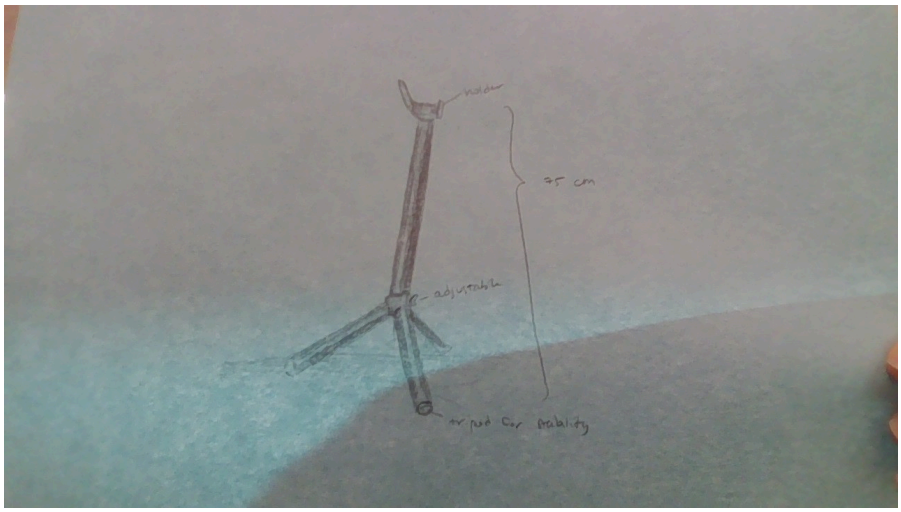
Date: 03/10/2025
Description: A wall rack that consists of multiple guitar hooks that can be hung on a wall and can hold multiple guitars.
Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____ **Date:** _____ **Class:** _____



Design Idea #3

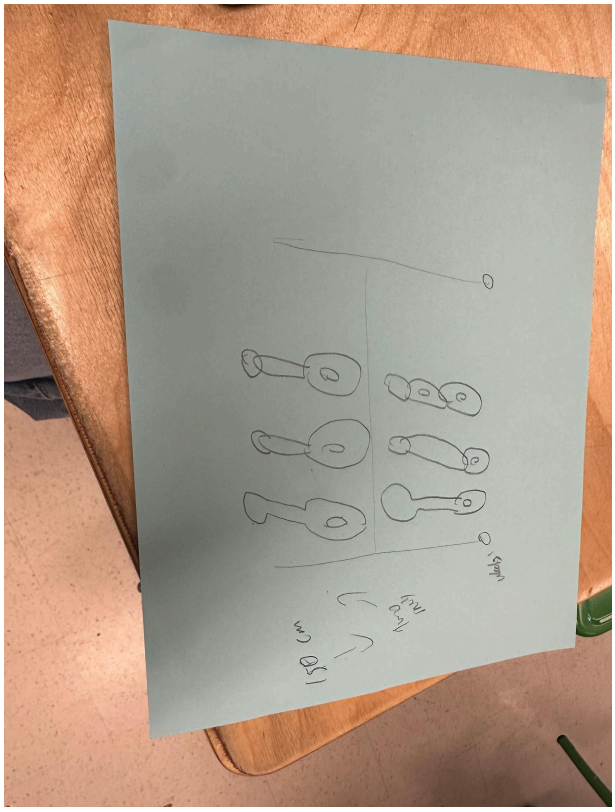
Date: 03/06/2025
Description: A stand for each individual guitar.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #4

Name: _____ **Date:** _____ **Class:** _____

Date: 03/06/2025
Description: A mobile rack that allows for multiple guitars to be stored on it.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection *(D1, D2)*

	weight	Closet	Wall Rack	Stand	Mobile Rack
Time	30%		1	1	
Portability	15%	1		1	1
Flexibility	15%	1		1	1
Cost	30%		1	1	

Name: _____ **Date:** _____ **Class:** _____

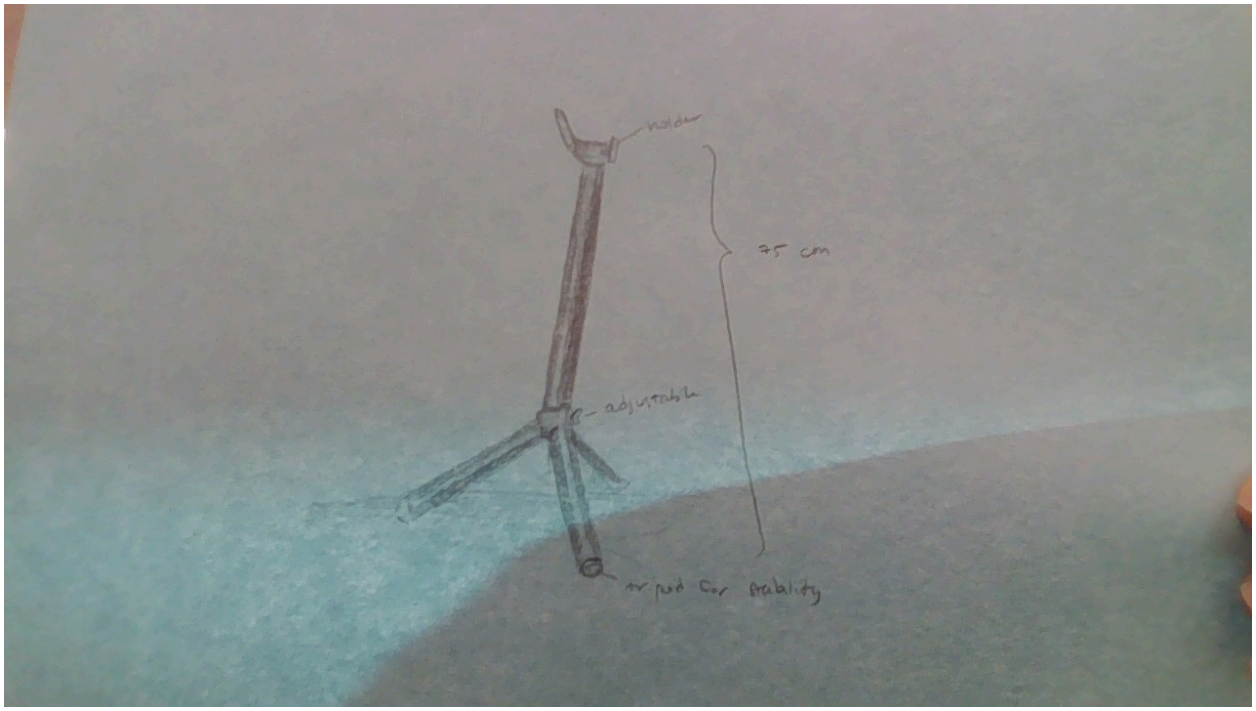
Creativity	10%	1	1		1
score	100%	3	3	4	3

Our team chooses design Stand **to solve the community problem.**

Design Blueprint *(D4)*

Design Blueprint #1

Date: 03/10/2025
Sketch:



Manufacturing Methods: 3D Printing, Cutting.
Step 1: Cut wood dowel into 4-5 pieces.
Step 2: Nail pieces together into desired shape
Step 3: Add 3D printed parts to wood structure.
Supply List and Potential Costs:

Name: _____ **Date:** _____ **Class:** _____

Item	**Cost each**	**# needed**	**Total**
Wood Dowel (1 in diameter)	$8	5	$40
Box of Screws	$6	5	$30
3D Printed Filament	$0	<0.1	$0
Total Cost	$14	5	$70

05/09/2025

We want to change our prototype by adding a wall mount and a 3D printed filament to it. This is because during the Designathon, Mr. Antonucci came up to our design and gave us some feedback, which included him saying that we should consider adding a wall mount. We are going to need access to a 3D printer at school or at someone's house.

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0

Name: _____ **Date:** _____ **Class:** _____

1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was *comprehensive, iterative, and consistently* defensible (viable design highly likely)	was *thorough, iterative, and* generally defensible (viable design likely)	was *adequate, generally* iterative and defensible *(viable design is possible)*	was *partial or overly general and only somewhat* iterative and/or defensible (raising issues with the design viability)	*was incomplete or minimally iterative and/or defensible (raising issues with the design viability)*	*was not evident that there was an attempt*
2.Design Solution Choice. The Design Solution chosen _____.	was *well-justified* and demonstrated attention to *all* design requirements	*was justified* and demonstrated attention to *most if not all* design requirements	*was explained with reference to at least some* design requirements	*was not sufficiently explained with reference to* design requirements	*lacked support related to* design requirements	*was not attempted or evidence not provided*
3. Plan of Action. The plan of action _____.	*has considerable merit and would easily* support repetition and testing for effectiveness by others	*would* support repetition and testing for effectiveness by others	*might not clearly or fully support repetition and testing for effectiveness by others*	*has insufficient detail to allow for testing for replication of results*	*is lacking to the point that the design solution cannot be tested*	*cannot be executed or does not exist*

Name: _____ **Date:** _____ **Class:** _____

May 5 2025

Our team decided that to change our prototype based on the feedback given, we would convert our stand from a floor stand to a wall rack. To do that, we would first remove the bottom legs of the stand. Afterwards, we would attach a hanger or an attachment to mount the stand to the wall. Finally, we would add another holder layer to the stand so it can hold a guitar effectively. Weiz and I are going to work on reconstructing the stand. WIlliam is tasked to gather the parts and supplies. Kyle's job is to keep track of our activities and record any data and progress we did. We plan to not purchase any materials and only use the materials we find in the classroom.